

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 11: Stall Warning & Angle of Attack Systems

Syllabus Source: CAE Oxford (pp. 399-438) | Pass Target: >= 90% AviationExam

1. ANGLE OF ATTACK (AOA) AERODYNAMICS & SENSOR TYPES

An airfoil stalls when airflow separates from the upper surface at the Critical Angle of Attack (α_{crit} , typically 14° to 18°). An aircraft stalls at the EXACT SAME CRITICAL AOA regardless of weight, airspeed, bank angle, or altitude:

- **Pivoting Aerodynamic Vanes:** Mounted on the left and right forward fuselage. A counterweighted, aerodynamically balanced wedge vane rotates freely to align with the local oncoming relative airflow. Electrically heated to prevent ice accumulation.
- **Conical / Slotted Probes (Rosemount):** A fixed cylindrical/conical probe projecting into the airstream with two longitudinal slots. Differential pressure between the two slots is measured by a internal transducer, providing AOA measurement with no external moving parts.

2. STALL WARNING & PREVENTION DEVICES: STICK SHAKER VS STICK PUSHER

CS-25 transport certification requires clear, unmistakable aerodynamic warning prior to stall entry:

System Component	Physical Mechanism & Trigger Point	Certification Rule & Aircraft Protection
System Component	Physical Mechanism & Trigger Point	Certification Rule & Aircraft Protection
Aerodynamic Buffet	Natural turbulent wake from stalled wing root buffeting horizontal stabilizer.	Light aircraft and low-wing jets. Some swept-wing jets lack natural pre-stall buffet, requiring artificial warning devices!
Stick Shaker (Artificial Warning)	Eccentric unbalanced electric motors mounted on the control columns vibrate the yoke vigorously at ~15 to 30 Hz with a loud clattering noise.	MANDATORY CS-25 requirement: Must activate at least 5 KNOTS OR 5% ABOVE the stall speed (V_S) to allow pilot recovery before the actual aerodynamic stall.
Stick Pusher (Artificial Prevention)	Pneumatic or hydraulic actuator ram connected to the elevator control linkage. Pushes the yoke violently FORWARD with 30 to 40 kg (60-80 lb) of force!	MANDATORY on T-tail swept-wing aircraft (e.g. CRJ, bizjets) susceptible to DEEP STALL (Super-Stall) where stalled wing wake blankets the T-tail elevator, making recovery aerodynamic impossible! Pushes nose down before critical AOA.

[!] EXAM TRAP (AVIATIONEXAM) - Deep Stall (Super-Stall) & Stick Pusher Role

- Deep Stall Risk: Swept-wing aircraft with T-tail empennage and rear-fuselage engines. - Phenomenon: At high angles of attack, wingtip stall causes center of lift to shift forward, pitching the nose UP further! The turbulent separated wake of the stalled wing blankets the high T-tail elevator, completely neutralizing pitch control! - Solution: The STICK PUSHER detects rapid approach to critical AOA and forcibly drives the elevator DOWN before deep stall can ever be entered!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Stall Warning System Hierarchy

Level 1: Angle of Attack Vane detects approach to stall. Level 2: Stick Shaker vibrates control column at 1.05 Vs (tactile warning). Level 3: Stick Pusher fires at critical stall angle, pushing column forward with 35-40 kg force to prevent unrecoverable T-tail deep stall.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA CS-25 certification requirements, at what speed margin must the artificial stall warning (stick

- [A]: Exactly at the aerodynamic stall speed
- [B]: At least 5 KNOTS or 5% ABOVE the stall speed (V_{S})
- [C]: 20 knots above stall speed
- [D]: Only when the landing gear is retracted

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CS-25 mandates that clear artificial or natural stall warning must begin at least 5 knots or 5% above the stall speed, providing sufficient reaction time for flight crew recovery.

Q2: Why are T-tail jet transport aircraft equipped with a mandatory STICK PUSHER system?

- [A]: To help the pilot trim the aircraft during take-off
- [B]: To prevent the aircraft from entering a catastrophic DEEP STALL, where the stalled wing wake blankets the T-tail, rendering elevators aerodynamically ineffective
- [C]: To retract the flaps automatically
- [D]: To prevent overspeed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In T-tail aircraft, wing wake washes over the elevated horizontal stabilizer at high angles of attack, causing complete loss of elevator pitch-down authority (deep stall). A stick pusher forcibly lowers the nose before this occurs.

Q3: Does an aircraft's critical stalling Angle of Attack change with increases in aircraft gross mass?

- [A]: Yes, critical AOA increases with mass
- [B]: No, the critical stalling Angle of Attack remains CONSTANT regardless of aircraft mass, bank angle, or altitude
- [C]: Yes, critical AOA decreases with mass
- [D]: It changes only with flap extension

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An airfoil stalls when the critical angle of attack is reached (airflow separation). While the stall SPEED increases with mass and load factor, the critical stalling ANGLE OF ATTACK is aerodynamically constant.